SET-A

Question-1: Arrange the following types of memory in order from slowest to fastest: a) DRAM, b) register, c) L2 cache, d) L3 cache, and e) disk.
Answer: e a d c b

Question-2: In an 8-bit operating system, at most how many process address spaces can be accommodated in 1 KB of DRAM without the use of demand paging: a) 4, b) 8, c) 16, d) 32?
Answer: a

Question-3: Calculate the cost of demand paging assuming DRAM access time as 100 nanoseconds, probability of page fault as 10%, and average page fault service time as 100 nanoseconds: a) 100ns, b) 110ns, c) 120ns, and d) 100.10ns.
Answer: b

Question-4: Which of the following strategies will lead to minimum overheads for parallelizing a Fibonacci number generator on a quad-core processor: a) using 4 processes, b) using 4 threads, c) using 8 processes, and d) using 8 threads?
Answer: **b**

Question-5: Which of the following parameters to pthread_create is incorrect:
pthread_create(int *thread /***A***/, const pthread_attr_t *a /***B***/, void (foo)() /***C***/, void* b /***D***/);
Answer: **A and C**

Question-6: Calculate the speedup when a sequential program of 12 seconds execution time consumes 6 seconds to execute when run in parallelized version on quad core processor: a) 4, b) 2, c) 1.5, and d) 6?
Answer: **b**

Question-7: What will be the parallel efficiency in the question above: a) 0.25, b) 0.5, c) 0.75, d) 1?
Answer: **b**

Question-8: Which of the following scheduling policy/policies optimize average completion time: a) FCFS, b) SJF, c) STCF, and d) RR?
Answer: **b and c**

Question-9: Which of the scheduling policy/policies mentioned in above question optimize average waiting time?
Answer: **d**

Question-10: Which of the following(s) will avoid deadlock: a) pthread_mutex_lock, b) pthread_mutex_trylock, c) circular wait, d) pthread_cond_signal?
Answer: **b**

Question-11: Which mechanism(s) ensures **only** one thread enters a critical section at a time: a) semaphore, b) mutex, c) condition variables, d) all of the above.
Answer: **b**

Question-12: In a 16-bit system using a 2-level page table and a page size of 64 bytes, how many bits are used for the page directory index (PDI), the page table index (PTI), and the offset (in same order, and assume equal number of bits for PDI and PTI): a) 4,4,8, b) 5,5,6, c) 3,3,10?
Answer: **b**

Question-13: Which of these register(s) is/are used in transforming the **logical** address to **virtual** address: a) PTBR, b) SS, c) DS, d) GDTR?
Answer: b c d

Question-14: Using the LRU page replacement policy and a page table with 2 frames, how many page faults occur for the following page reference string: "**A, B, C, D, D, B, A, B, D**" a) 5, b) 6, c) 7, d) 4.
Answer: **c**

SET-B

Question-1: Arrange the following types of memory in order from slowest to fastest: a) register, b) DRAM, c) L3 cache, d) L2 cache, and e) disk.
Answer: e b c d a

Question-2: In an 8-bit operating system, at most how many process address spaces can be accommodated in 1 KB of DRAM without the use of demand paging: a) 8, b) 4, c) 16, d) 32?
Answer: b

Question-3: Calculate the cost of demand paging assuming DRAM access time as 100 nanoseconds, probability of page fault as 10%, and average page fault service time as 100 nanoseconds: a) 110ns, b) 100ns, c) 120ns, and d) 100.10ns.
Answer: a

Question-4: Which of the following strategies will lead to minimum overheads for parallelizing a Fibonacci number generator on a quad-core processor: a) using 4 processes, b) using 8 threads, c) using 8 processes, and d) using 4 threads?
Answer: **d**

Question-5: Which of the following parameters to pthread_create is incorrect:
pthread_create(int *thread /***A***/, const pthread_attr_t *a /***B***/, void (foo)() /***C***/, void* b /***D***/);
Answer: **A and C**

Question-6: Calculate the speedup when a sequential program of 12 seconds execution time consumes 6 seconds to execute when run in parallelized version on quad core processor: a) 4, b) 1.5, c) 2, and d) 6?
Answer: **c**

Question-7: What will be the parallel efficiency in the question above: a) 0.5, b) 0.25, c) 0.75, d) 1?
Answer: **a**

Question-8: Which of the following scheduling policy/policies optimize average completion time: a) SJF, b) FCFS, c) STCF, and d) RR?
Answer: **a and c**

Question-9: Which of the scheduling policy/policies mentioned in above question optimize average waiting time?
Answer: **d**

Question-10: Which of the following(s) will avoid deadlock: a) pthread_mutex_lock, b) pthread_mutex_trylock, c) circular wait, d) pthread_cond_signal?
Answer: **b**

Question-11: Which mechanism(s) ensures **only** one thread enters a critical section at a time: a) semaphore, b) mutex, c) condition variables, d) all of the above.
Answer: **b**

Question-12: In a 16-bit system using a 2-level page table and a page size of 64 bytes, how many bits are used for the page directory index (PDI), the page table index (PTI), and the offset (in same order, and assume equal number of bits for PDI and PTI): a) 4,4,8, b) 3,3,10 c) 5,5,6?
Answer: **c**

Question-13: Which of these register(s) is/are used in transforming the **logical** address to **virtual** address: a) PTBR, b) SS, c) DS, d) GDTR?
Answer: b c d

Question-14: Using the LRU page replacement policy and a page table with 2 frames, how many page faults occur for the following page reference string: "**A, B, C, D, D, B, A, B, D**" a) 7, b) 6, c) 4, d) 5.
Answer: **a**

SET-C

Question-1: Which of the following strategies will lead to minimum overheads for parallelizing a Fibonacci number generator on a quad-core processor: a) using 4 processes, b) using 8 threads, c) using 8 processes, and d) using 4 threads?
Answer: **d**

Question-2: Calculate the speedup when a sequential program of 12 seconds execution time consumes 6 seconds to execute when run in parallelized version on quad core processor: a) 4, b) 1.5, c) 2, and d) 6?
Answer: **c**

Question-3: Calculate the cost of demand paging assuming DRAM access time as 100 nanoseconds, probability of page fault as 10%, and average page fault service time as 100 nanoseconds: a) 110ns, b) 100ns, c) 120ns, and d) 100.10ns.
Answer: a

Question-4: Arrange the following types of memory in order from slowest to fastest: a) register, b) DRAM, c) L3 cache, d) L2 cache, and e) disk.
Answer: e b c d a

Question-5: Which of the following parameters to pthread_create is incorrect:
pthread_create(int *thread /***A***/, const pthread_attr_t *a /***B***/, void (foo)() /***C***/, void* b /***D***/);
Answer: **A and C**

Question-6: In an 8-bit operating system, at most how many process address spaces can be accommodated in 1 KB of DRAM without the use of demand paging: a) 8, b) 4, c) 16, d) 32?
Answer: b

Question-7: What will be the parallel efficiency in the Question-2 above: a) 0.25, b) 0.75, c) 0.5, d) 1?
Answer: **c**

Question-8: Which of the following scheduling policy/policies optimize average completion time: a) SJF, b) FCFS, c) STCF, and d) RR?
Answer: **a and c**

Question-9: Which of the scheduling policy/policies mentioned in above question optimize average waiting time?
Answer: **d**

Question-10: Which of the following(s) will avoid deadlock: a) pthread_mutex_trylock, b) pthread_mutex_lock, c) circular wait, d) pthread_cond_signal?
Answer: **a**

Question-11: Which mechanism(s) ensures **only** one thread enters a critical section at a time: a) semaphore, b) mutex, c) condition variables, d) all of the above.
Answer: **b**

Question-12: In a 16-bit system using a 2-level page table and a page size of 64 bytes, how many bits are used for the page directory index (PDI), the page table index (PTI), and the offset (in same order, and assume equal number of bits for PDI and PTI): a) 4,4,8, b) 3,3,10 c) 5,5,6?
Answer: **c**

Question-13: Which of these register(s) is/are used in transforming the **logical** address to **virtual** address: a) PTBR, b) SS, c) DS, d) GDTR?
Answer: b c d

Question-14: Using the LRU page replacement policy and a page table with 2 frames, how many page faults occur for the following page reference string: "**A, B, C, D, D, B, A, B, D, C**" a) 7, b) 6, c) 4, d) 8.
Answer: **d**